

Experimental Evaluation and Optimization of Real-Time Scheduling on STM32

Vladimir Toporkov
Computer Science and Engineering
Innopolis University
Innopolis, Russia
v.toporkov@innopolis.university

ToDo

I. INTRODUCTION

A. Contribution

We propose the following contribution:

- Experimental evaluation and comparison of Super Loop and Earliest Deadline First on STM32.

II. BACKGROUND ON STM32

A typical STM32 system consists of two components:

- 1) Hardware - STM32 microcontroller (uniprocessor) and sensors.
- 2) Software - sensor logic and scheduler, with Super Loop used as the default baseline.

A. STM32 hardware

An example of the hardware system is shown in Fig. 1.

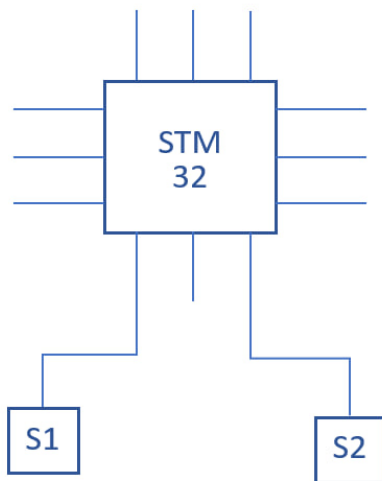


Fig. 1. Hardware components of the STM32 system (S_1 , S_2 - sensors).

B. STM32 scheduling

P.

Super Loop does not account for job deadlines, baseline profiling shows deadline misses and skipped releases under different workload configurations, as shown in Fig. 3.

Earliest Deadline First (EDF) as a possible solution accounts for job deadlines, but its performance on STM32 requires experimental evaluation.

The goal is to evaluate schedulability on STM32, identify its main bottlenecks, and improve the scheduling behavior.

This leads to several research ideas:

- 1) Compare EDF and Super Loop scheduling under the same task set (**key metrics:** deadline-miss ratio, skipped-release ratio, response time).
- 2) Improve EDF behavior under high CPU utilization using feedback control (**key metrics:** deadline-miss ratio, skipped-release ratio, response time, CPU utilization).
- 3) Minimize unnecessary context switches in EDF (**key metric:** number of preemptions).

III. BACKGROUND ON SCHEDULING

A. Superloop Scheduler: Key Principles

B. EDF

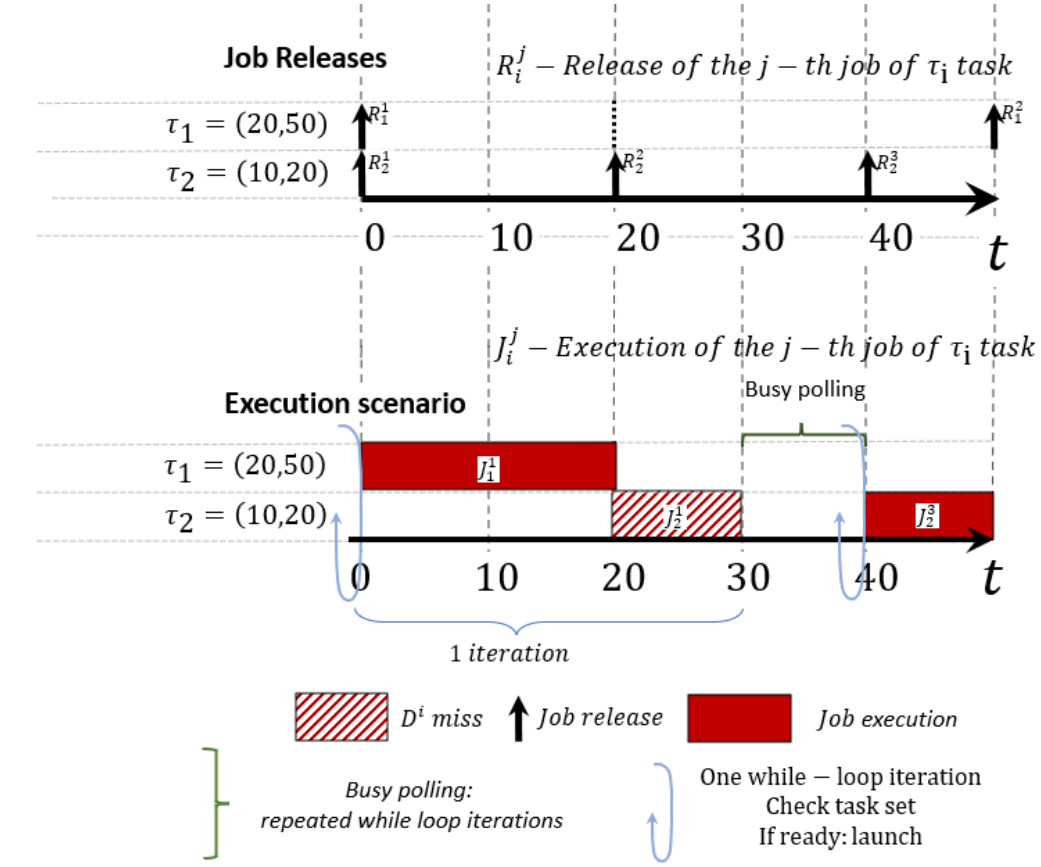
ToDo

C. Brief Comparison

ToDo (Table)

IV. SCHEDULERS IMPLEMENTATION FOR STM32

ToDo (code?)



next release time of task τ_i current system time

t^{cur}	0	20	30	40	50
t_1^{next}	0	50	50	50	50
t_2^{next}	0	0	40	40	60
miss? $\rightarrow t^{cur} > R_i^k + D^i$	–	$20 < R_1^1 + 50$	$30 > R_2^1 + 20$	–	$50 < R_2^3 + 20$
skip? $\rightarrow t^{cur} > t_i^{next}$	–	$20 < R_1^1$	$30 > R_2^2$ $30 < R_2^3$	–	$50 < R_2^4$

Fig. 2. Super Loop execution scenario and corresponding scheduling state.

V. PERFORMANCE EVALUATION

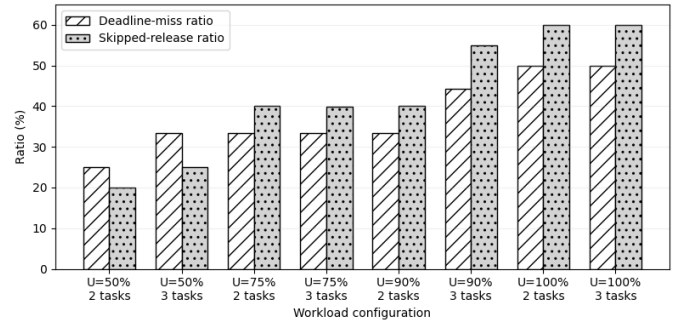


Fig. 3. Super Loop deadline-miss and skipped-release ratios for a specific task under different CPU utilization (U) levels and task-set sizes.